TM Forum Specification

Service Usage Management User Guide

TMF727

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Introduction

This document is the user-guide of the REST API for Service Usage Management. It includes the data model definition as well as all available operations for the ServiceUsage and ServiceUsageSpecification resources.

The ServiceUsage API provides a standardized mechanism for service usage management such as the creation, update, retrieval, import and export of a collection of service usages. A service usage is an occurrence of usage on a Service derived from various Resource usages, which can be further used by BSS for determining the Product usage.

It is comprised of characteristics, which represent attributes of service usage.

The ServiceUsage API performs the following operations on the ServiceUsage resource:
* Retrieval of a ServiceUsage or a collection of ServiceUsage, depending on filter criteria
* Partial update of a ServiceUsage or of a collection of ServiceUsage
* Creation of a ServiceUsage

The ServiceUsageSpecification API provides a standardized mechanism for service usage specification management such as the creation, update, retrieval of a service usage specification. Service Usage Specification defines the detailed description of a service usage event that are of interest to the business. It is comprised of characteristics, which define all attributes known for a particular type of usage.

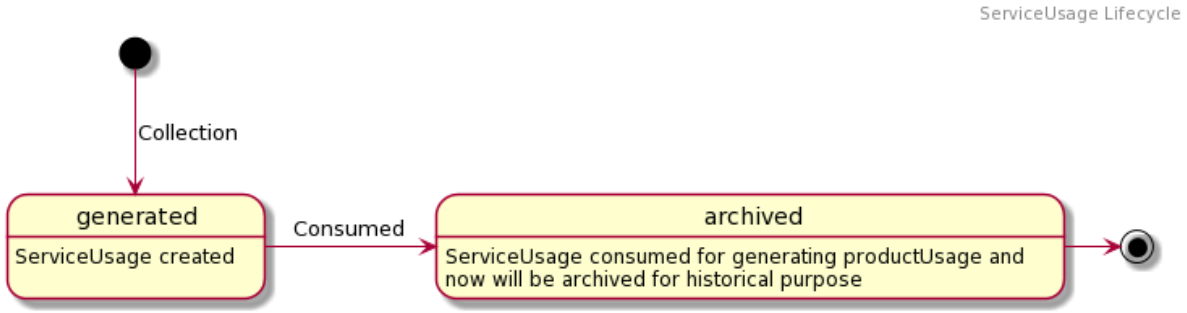
The ServiceUsageSpecification API perform the following operations on the ServiceUsageSpecification resource:
* Retrieval of a ServiceUsageSpecification or a collection of ServiceUsageSpecification, depending on filter criteria
* Partial update of a ServiceUsageSpecification or of a collection of ServiceUsageSpecification
* Creation of a ServiceUsageSpecification

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Sample Use Cases

A few simple use-cases are presented:
• Creation of a simple ServiceUsageSpecification, defining the characteristics of a voice call (caller and receipient numbers, start and stop date-and-time); listing ServiceUsageSpecification's ; retrieving a single ServiceUsageSpecification and ultimately its deletion based on the same id.
• Creation of a simple Usage relating to a voice call; listing Usage of serviceusageType "Voice"; retrieving a specific Usage based on it's id; and archiving the service usage based this Usage based on the same id; then ultimately deleting the service usage based on the same id
• Creation of a simple Usage relating to a voice mail retrieval; listing Usage of serviceusageType "Voicemail Retrieval"; retrieving a specific Usage based on it's id; and archiving the service usage based this Usage based on the same id; then ultimately deleting the service usage based on the same id
• Creation of a simple service Usage relating to a duration when a service instance or a feature is in active state (e.g. user activated an internet service for 30 minutes, user activated speed boost feature) to enable a consumption based service usage pattern; retrieving a specific Usage based on it's id; archiving the service usage based this Usage based on the same id ; then ultimately deleting the service usage based on the same id

ServiceUsage Life Cycle



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Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance a BillingAccount and SettlementAccount inheriting properties from the abstract Account entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines, Part 2 (TMF630).

The @type attribute provides a way to represent the actual class type of an entity. For example, within a list of Account instances some may be instances of BillingAccount where other could be instances of SettlementAccount. The @type gives this information. All resources and sub-resources of this API have a @type attributes that can be provided when this is useful.

The @referredType can be used within reference entities (like for instance an AccountRef object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the @type, when used, denotes the class type of the reference itself, such as BillingAccountRef or SettlementAccountRef, and not the class type of the referred object. However since reference classes are rarely sub-classed, @type is generally not useful in reference objects.

The @schemaLocation property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected characteristics of an entity.

The @baseType attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

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RESOURCE MODEL

Managed Entity and Task Resource Models

ServiceUsage resource

An occurrence of usage on a Service derived from various resource usages. It is comprised of characteristics, which represent attributes of service usage.

Resource model

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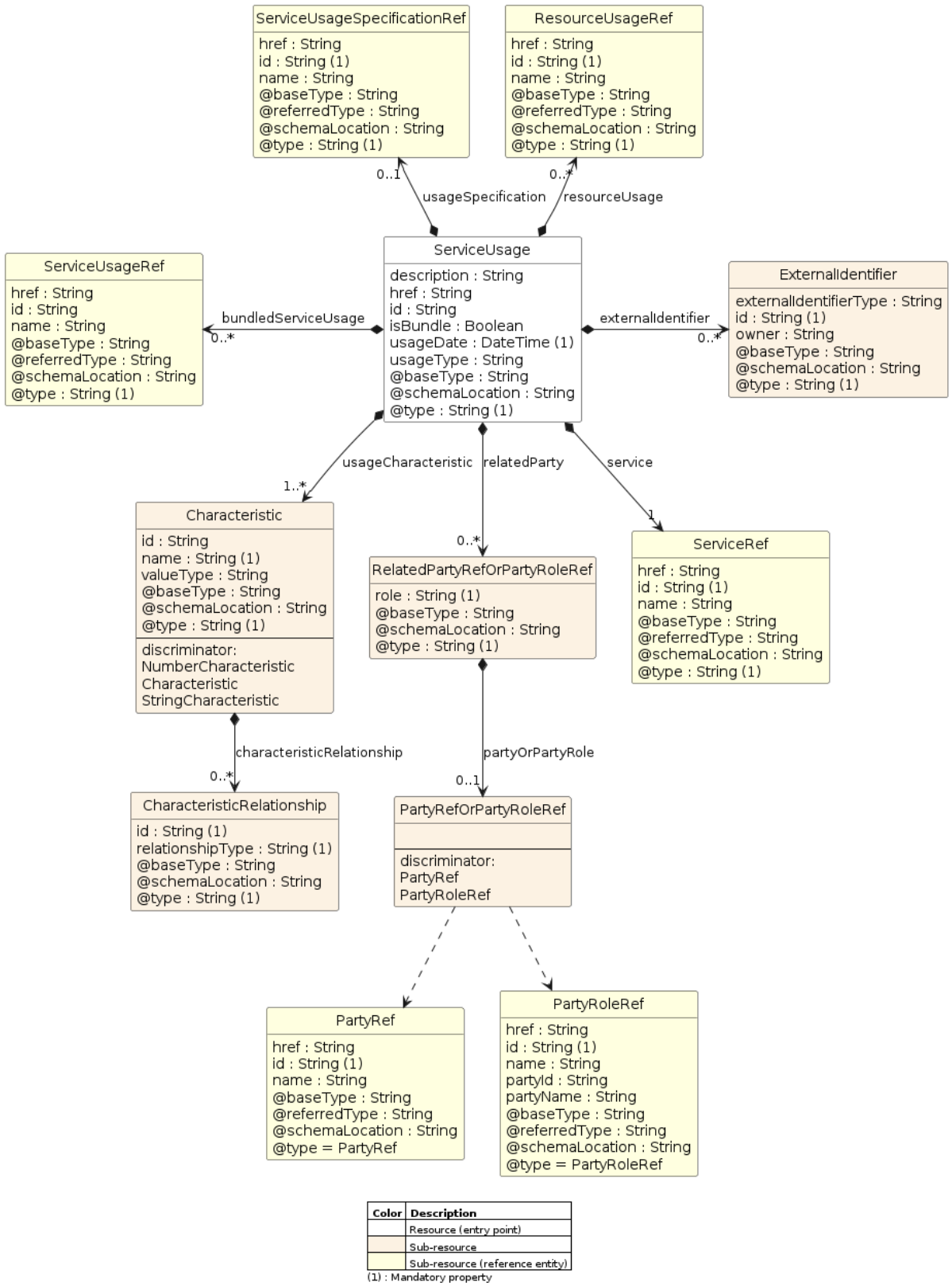


Figure 1 - ServiceUsage

Field descriptions

ServiceUsage fields

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bundledServiceUsage	A ServiceUsageRef. Service Usage reference. An occurrence of usage on a Service derived from various resource usages.
description	A String. Description of service usage.
externalIdentifier	An ExternalIdentifier. An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
isBundle	A Boolean. IsBundle determines whether usage represents a single usage (false), or a bundle of usages specifications (true).
relatedParty	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.
resourceUsage	A ResourceUsageRef. ResourceUsage reference. ResourceUsage is usage event for Resource.
service	A ServiceRef. Service reference, for when Service is used by other entities.
usageCharacteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic… Characteristic can be instanciated as * NumberCharacteristic * StringCharacteristic
usageDate	A DateTime. Date of usage.
usageSpecification	A ServiceUsageSpecificationRef. UsageSpecification reference. UsageSpecification is a detailed description of a service usage event that are of interest to the business. It is comprised of characteristics, which define all attributes known for a particular type of usage.
usageType	A String. Type of usage.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String. When sub-classing, this defines the sub-class Extensible name.

ServiceUsageRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. The name of the usage.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ExternalIdentifier sub-resource fields

externalIdentifierType	A String. Type of the identification, typically would be the type of the entity within the external system.
id	A String. Identification of the entity within the external system.
owner	A String. Name of the external system that owns the entity.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ServiceUsageSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Characteristic sub-resource fields

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characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	Characteristic can be instanciated as * NumberCharacteristic * StringCharacteristic
value	This property is present in subclasses

NumberCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Number. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A String. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPartyRefOrPartyRoleRef sub-resource fields

partyOrPartyRole	A PartyRefOrPartyRoleRef.
role	A String. Role played by the related party or party role in the context of the specific entity it is linked to. Such as 'initiator', 'customer', 'salesAgent', 'user'.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceUsageRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ServiceRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CharacteristicRelationship sub-resource fields

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id	A String. Unique identifier of the characteristic.
relationshipType	A String. The type of relationship.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRoleRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
partyId	A String. The identifier of the engaged party that is linked to the PartyRole object.
partyName	A String. The name of the engaged party that is linked to the PartyRole object.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'ServiceUsage' resource object.

```
{
```

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```
    "id": "5130d5131a23",
    "href":
"https://api.csp.com/serviceUsageManagement/v5/serviceUsage/5130d5131a23",
    "usageDate": "2020-11-20T16:07:00Z",
    "description": "Voice Call",
    "usageType": "Voice",
    "status": "generated",
    "@type": "ServiceUsage",
    "service": {
        "id": "8132-b246a6746137",
        "href": "https://api.csp.com/serviceInventory/v5/service/8132-b246a6746137",
        "name": "Ivor voice service",
        "@type":"Service"
    },
    "usageSpecification": {
        "id": "8b64-0cfd1a90c9ed",
        "href":
"https://api.csp.com/serviceUsageManagement/v5/serviceUsageSpecification/8b64-
0cfd1a90c9ed",
        "name": "Voice Call Usage Specification",
        "@referredType": "ServiceUsageSpecification",
        "@type":"ServiceUsageSpecificationRef"
    },
    "relatedParty": [
        {
            "role": "Customer",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "id": "8185-b246a6746136",
              "href":
"https://serverlocation:port/partyManagement/v5/individual/8185-b246a6746136",
              "name": "Ivor Message",
              "@referredType": "Individual"
            }
        }
    ],
    "usageCharacteristic": [
        {
            "name": "originatingCountryCode",
            "value": "43",
            "valueType": "string",
            "@type":"StringCharacteristic"
        },
        {
            "name": "originatingNumber",
            "value": "676123456789",
            "valueType": "string",
            "@type":"StringCharacteristic"
        },
        {
            "name": "destinationCountryCode",
            "value": "49",
            "valueType": "string",
            "@type":"StringCharacteristic"
        },
        {
            "name": "destinationNumber",
            "value": "170123456789",
            "valueType": "string",
            "@type":"StringCharacteristic"
        },
        {
```


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```
            "name": "duration",
            "value": 20,
            "valueType": "number",
            "@type":"NumberCharacteristic"
        },
        {
            "name": "unit",
            "value": "SEC",
            "valueType": "string",
            "@type":"StringCharacteristic"
        }
    ]
}
```


ServiceUsageSpecification resource

A detailed description of a service usage event that are of interest to the business. It is comprised of characteristics, which define all attributes known for a particular type of usage.

Resource model

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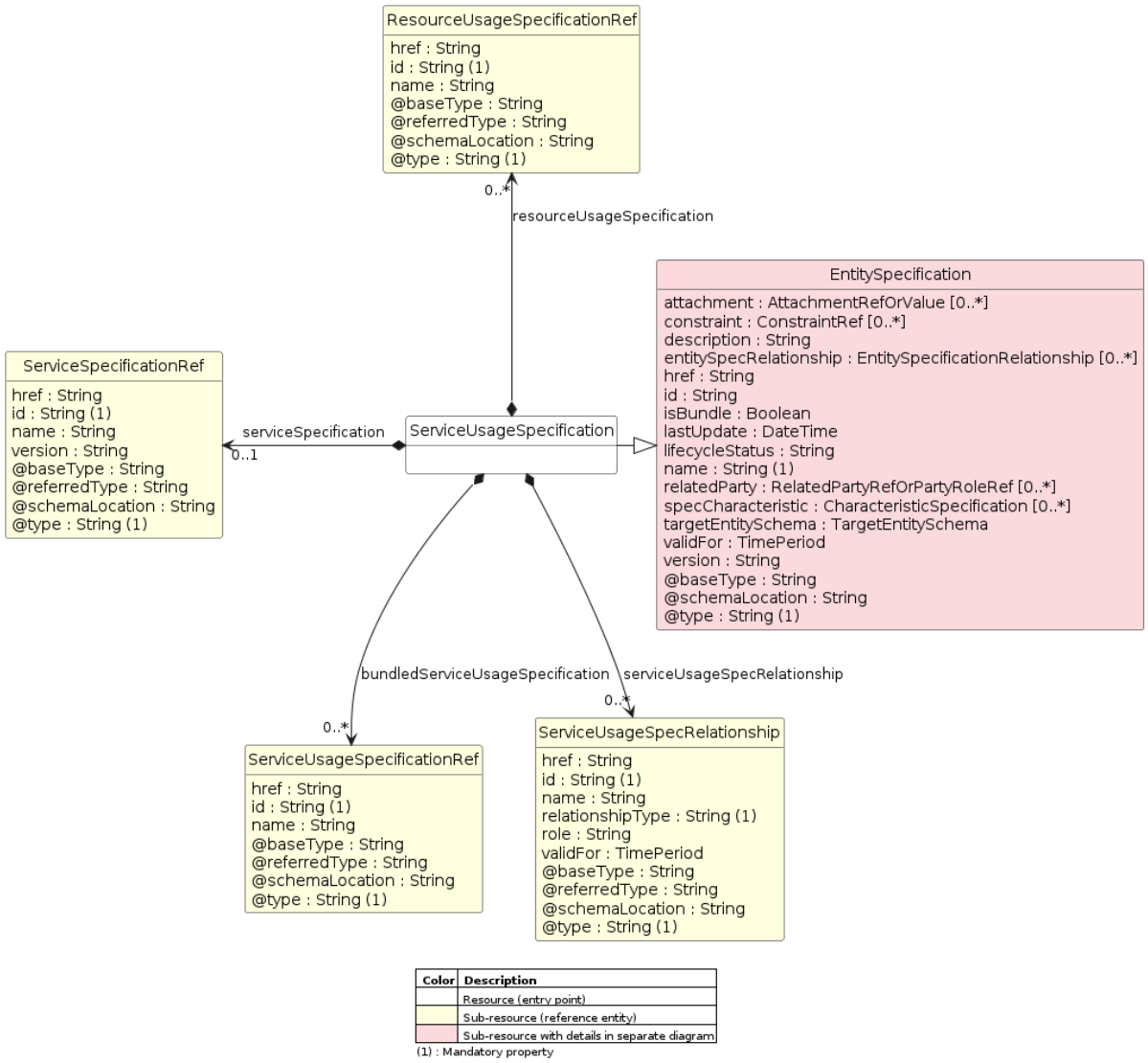


Figure 2 - ServiceUsageSpecification

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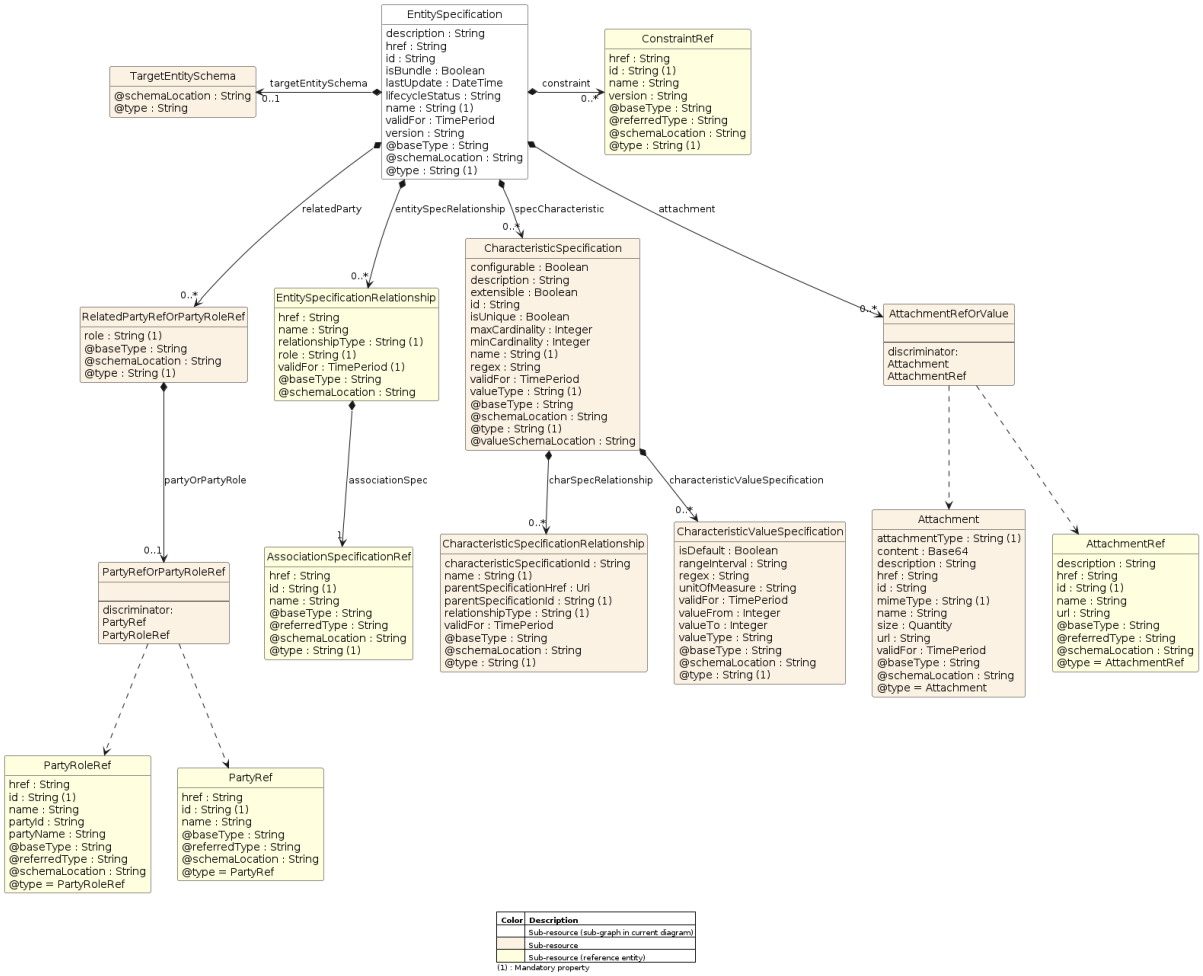


Figure 3 - EntitySpecification

Field descriptions

ServiceUsageSpecification fields

attachment	An AttachmentRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Attachment entity and not the AttachmentRefOrValue class itself.
bundledServiceUsageSpecification	A ServiceUsageSpecificationRef. UsageSpecification reference. UsageSpecification is a detailed description of a service usage event that are of interest to the business. It is comprised of characteristics, which define all attributes known for a particular type of usage.
constraint	A ConstraintRef. Constraint reference. The Constraint resource represents a policy/rule applied to an entity or entity spec.
description	A String. Description of the specification.
entitySpecRelationship	An EntitySpecificationRelationship. A migration, substitution, dependency or exclusivity relationship between/among entity specifications.
href	A String. Hyperlink reference.

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id	A String. Unique identifier.
isBundle	A Boolean. IsBundle determines whether specification represents a single specification (false), or a bundle of specifications (true).
lastUpdate	A DateTime. Date and time of the last update of the specification.
lifecycleStatus	A String. Used to indicate the current lifecycle status of this catalog item.
name	A String. Name given to the specification.
relatedParty	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.
resourceUsageSpecification	A ResourceUsageSpecificationRef. ResourceUsageSpecification reference. ResourceUsageSpecification is a detailed description of a usage event that are of interest to the business and can have charges applied to it. It is comprised of characteristics, which define all attributes known for a particular type of usage.
serviceSpecification	A ServiceSpecificationRef. Service specification reference: ServiceSpecification(s) required to realize a ProductSpecification.
serviceUsageSpecRelationship	A ServiceUsageSpecRelationship. A dependency, substitution or exclusivity relationship between/among service usage specifications.
specCharacteristic	A CharacteristicSpecification. This class defines a characteristic specification.
targetEntitySchema	A TargetEntitySchema. The reference object to the schema and type of target entity which is described by a specification.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
version	A String. Specification version.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ServiceSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String. Service specification version.
@baseType	A String. When sub-classing, this defines the super-class.

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@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceUsageSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ServiceUsageSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ServiceUsageSpecRelationship sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
relationshipType	A String. Type of relationship such as dependency, substitution or exclusivity.
role	A String. The association role for this service specification.

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validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'ServiceUsageSpecification' resource object.

```
{
    "id": "8b64-0cfd1a90c9ed",
    "href":
"https://api.csp.com/serviceUsageManagement/v5/serviceUsageSpecification/8b64-
0cfd1a90c9ed",
    "description": "Specification for voice calls usage",
    "name": "Voice-Call-Specification",
    "isBundle": true,
    "bundledServiceUsageSpecification" : [
        {   "id": "90c9ed-8b64",
            "@type":"ServiceUsageSpecification",
            "name": "Local calls"
        },
        {   "id": "435d-87884",
            "@type":"ServiceUsageSpecification",
            "name": "Long-dinstance calls"
        }
    ],
    "specCharacteristic": [
        {
            "name": "originatingCountryCode",
            "description": "country code of the caller",
            "configurable": true,
            "valueType": "number",
            "minCardinality": 1,
            "maxCardinality": 1,
            "isUnique": true,
            "@type":"CharacteristicSpecification",
            "extensible": true,
            "validFor": {
                "startDateTime": "2020-09-23T16:42:23Z"
            }
        },
        {
            "name": "originatingNumber",
            "description": "phone number of the caller",
            "configurable": true,
            "valueType": "number",
            "minCardinality": 1,
            "maxCardinality": 1,
            "isUnique": true,
```

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```
            "@type":"CharacteristicSpecification",
            "extensible": true,
            "validFor": {
                "startDateTime": "2020-09-23T16:42:23Z"
            }
        },
        {
            "name": "destinationCountryCode",
            "description": "country code of the called party",
            "configurable": true,
            "valueType": "number",
            "minCardinality": 1,
            "maxCardinality": 1,
            "isUnique": true,
            "@type":"CharacteristicSpecification",
            "extensible": true,
            "validFor": {
                "startDateTime": "2020-09-23T16:42:23Z"
            }
        },
        {
            "name": "destinationNumber",
            "description": "phone number of the called party",
            "configurable": true,
            "valueType": "number",
            "minCardinality": 1,
            "maxCardinality": 1,
            "isUnique": true,
            "@type":"CharacteristicSpecification",
            "extensible": true,
            "validFor": {
                "startDateTime": "2020-09-23T16:42:23Z"
            }
        },
        {
            "name": "duration",
            "description": "duration of the call",
            "configurable": true,
            "valueType": "number",
            "minCardinality": 1,
            "maxCardinality": 1,
            "@type":"CharacteristicSpecification",
            "extensible": true,
            "validFor": {
                "startDateTime": "2020-09-23T16:42:23Z"
            }
        },
        {
            "name": "unit",
            "description": "unit of the duration",
            "configurable": true,
            "valueType": "string",
            "minCardinality": 0,
            "maxCardinality": 1,
            "characteristicValueSpecification": [
                {
                    "isDefault": true,
                    "value": "SEC",
                    "@type":"CharacteristicValueSpecification"
                }
            ],
            "@type":"CharacteristicSpecification",
            "extensible": true,
```


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```
            "validFor": {
                "startDateTime": "2020-09-23T16:42:23Z"
            }
        },
        {
            "name": "startDateTime",
            "description": "startdate and starttime of the call",
            "configurable": true,
            "valueType": "dateTime",
            "minCardinality": 0,
            "maxCardinality": 1,
            "@type":"CharacteristicSpecification",
            "extensible": true,
            "validFor": {
                "startDateTime": "2020-09-23T16:42:23Z"
            }
        },
        {
            "name": "endDateTime",
            "description": "enddate and endtime of the call",
            "configurable": true,
            "valueType": "dateTime",
            "minCardinality": 0,
            "maxCardinality": 1,
            "@type":"CharacteristicSpecification",
            "extensible": true,
            "validFor": {
                "startDateTime": "2020-09-23T16:42:23Z"
            }
        }
    ],
    "validFor": {
        "startDateTime": "2022-02-20T17:00:00.000Z",
        "endDateTime": "2099-01-11T00:00:00.000Z",
        "@type":"TimePeriod"
    },
    "@type":"ServiceUsageSpecification"
}
```


Notification Resource Models

7 notifications are defined for this API.

Notifications related to ServiceUsage:

- Create Event
- Delete Event
- Attribute Value Change Event

Notifications related to ServiceUsageSpecification:

- Create Event
- Delete Event
- Attribute Value Change Event
- State Change Event

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The notification structure for all notifications in this API follow the pattern depicted by the figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType). This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

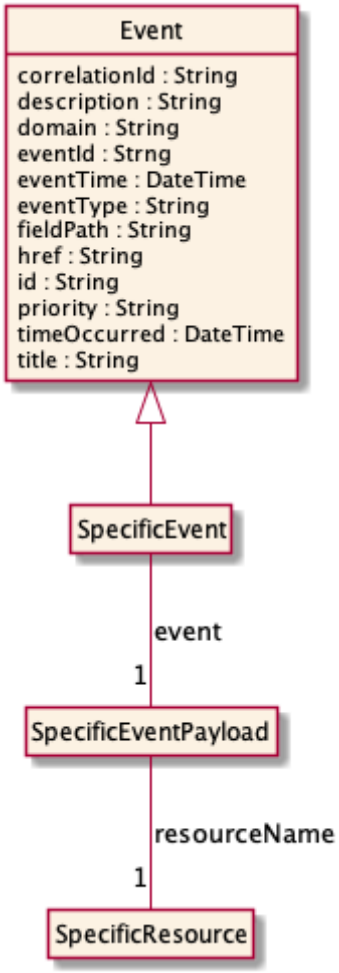


Figure 4 Notification Pattern

ServiceUsage

Create Event

Message example for ServiceUsageCreateEvent event

```
Content-Type: application/json

{
    "correlationId": "30517de1-15f0",
    "description": "ServiceUsageCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "4234-baa9-5e9e279f8174",
    "eventTime": "2023-03-06T04:18:39.349Z",
    "eventType": "ServiceUsageCreateEvent",
```

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```
    "priority": "3",
    "title": "ServiceUsageCreateEvent",
    "event": {
      "serviceUsage": {
        "id": "8b64-0cfd1a90c9ed",
        "href": "https://api.csp.com/serviceUsageManagement/v5/serviceUsage/8b64-
0cfd1a90c9ed",
        "description": "Voice Call",
        "usageType": "Voice",
        "service": {
          "id": "8132-b246a6746137",
          "href": "https://api.csp.com/serviceInventory/v5/service/8132-
b246a6746137",
          "name": "Ivor voice service",
          "@type":"ServiceRef"
        },
        "usageSpecification": {
          "id": "8b64-0cfd1a90c9ed",
          "href":
"https://api.csp.com/serviceUsageManagement/v5/serviceUsageSpecification/8b64-
0cfd1a90c9ed",
          "name": "Voice Call Usage Specification",
          "@referredType": "ServiceUsageSpecification",
          "@type":"ServiceUsageSpecificationRef"
        },
        "relatedParty": [
          {
            "role": "Customer",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "id": "8185-b246a6746136",
              "href":
"https://serverlocation:port/partyManagement/v5/individual/8185-b246a6746136",
              "name": "Ivor Message",
              "@referredType": "Individual"
            }
          }
        ],
        "usageCharacteristic": [
          {
            "name": "originatingCountryCode",
            "value": "90",
            "valueType": "string",
            "@type":"StringCharacteristic"
          },
          {
            "name": "originatingNumber",
            "value": "676123456789",
            "valueType": "string",
            "@type":"StringCharacteristic"
          },
          {
            "name": "destinationCountryCode",
            "value": "49",
            "valueType": "string",
            "@type":"StringCharacteristic"
          },
          {
            "name": "destinationNumber",
            "value": "170123456789",
            "valueType": "string",
            "@type":"StringCharacteristic"
```


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```
          },
          {
            "name": "duration",
            "value": 20,
            "valueType": "number",
            "@type":"NumberCharacteristic"
          },
          {
            "name": "unit",
            "value": "SEC",
            "valueType": "string",
            "@type":"StringCharacteristic"
          }
        ],
        "@type":"ServiceUsage"
      }
    },
    "reportingSystem": {
      "id": "504",
      "name": "APP-302",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "source": {
      "id": "732",
      "name": "APP-641",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "ServiceUsageCreateEvent"
  }
```


Delete Event

Message example for ServiceUsageDeleteEvent event


```
Content-Type: application/json

{
    "correlationId": "17b26b4d-cfe4",
    "description": "ServiceUsageDeleteEvent illustration",
    "domain": "Commercial",
    "eventId": "4f3f-8fd6-67018c1f8b65",
    "eventTime": "2023-03-06T04:18:39.350Z",
    "eventType": "ServiceUsageDeleteEvent",
    "priority": "2",
    "title": "ServiceUsageDeleteEvent",
    "event": {
      "serviceUsage": {
        "id": "8b64-0cfd1a90c9ed",
        "href": "https://api.csp.com/serviceUsageManagement/v5/serviceUsage/8b64-
0cfd1a90c9ed",
        "@type":"ServiceUsage"
      }
    },
    "reportingSystem": {
      "id": "504",
      "name": "APP-302",
      "@type": "ReportingResource",
```


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```
      "@referredType": "LogicalResource"
    },
    "source": {
      "id": "732",
      "name": "APP-641",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "ServiceUsageDeleteEvent"
  }
```

Attribute Value Change Event

Message example for ServiceUsageAttributeValueChangeEvent event

```
Content-Type: application/json

{
    "correlationId": "30517de1-15f0",
    "description": "ServiceUsageAttributeValueChangeEvent illustration",
    "domain": "Commercial",
    "eventId": "4234-baa9-5e9e279f8174",
    "eventTime": "2023-03-06T04:18:39.349Z",
    "eventType": "ServiceUsageAttributeValueChangeEvent",
    "priority": "3",
    "title": "ServiceUsageAttributeValueChangeEvent",
    "event": {
      "serviceUsage": {
        "id": "8b64-0cfd1a90c9ed",
        "href": "https://api.csp.com/serviceUsageManagement/v5/serviceUsage/8b64-
0cfd1a90c9ed",
        "@type":"ServiceUsage",
        "externalIdentifier": [
          {
            "id": "usage-0001-009",
            "@type":"ExternalIdentifier",
            "externalIdentifierType": "UsageEntity"
          }
        ]
      }
    },
    "reportingSystem": {
      "id": "504",
      "name": "APP-302",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "source": {
      "id": "732",
      "name": "APP-641",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "ServiceUsageAttributeValueChangeEvent"
  }
```

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ServiceUsageSpecification

Create Event

Message example for ServiceUsageSpecificationCreateEvent event

```
Content-Type: application/json

{
    "correlationId": "30517de1-15f0",
    "description": "ServiceUsageSpecificationCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "4234-baa9-5e9e279f8174",
    "eventTime": "2023-03-06T04:18:39.349Z",
    "eventType": "ServiceUsageSpecificationCreateEvent",
    "priority": "3",
    "title": "ServiceUsageSpecificationCreateEvent",
    "event": {
        "serviceUsageSpecification": {
            "id": "8b64-0cfd1a90c9ed",
            "href":
"https://api.csp.com/serviceUsageManagement/v4/serviceUsageSpecification/8b64-
0cfd1a90c9ed",
            "description": "Specification for voice calls usage",
            "lifecycleStatus": "active",
            "name": "Voice-Call-Specification",
            "specCharacteristic": [
                {
                    "name": "originatingCountryCode",
                    "description": "country code of the caller",
                    "configurable": true,
                    "valueType": "number",
                    "minCardinality": 1,
                    "maxCardinality": 1,
                    "isUnique": true,
                    "@type":"CharacteristicSpecification"
                },
                {
                    "name": "originatingNumber",
                    "description": "phone number of the caller",
                    "configurable": true,
                    "valueType": "number",
                    "minCardinality": 1,
                    "maxCardinality": 1,
                    "isUnique": true,
                    "@type":"CharacteristicSpecification"
                },
                {
                    "name": "destinationCountryCode",
                    "description": "country code of the called party",
                    "configurable": true,
                    "valueType": "number",
                    "minCardinality": 1,
                    "maxCardinality": 1,
                    "isUnique": true,
                    "@type":"CharacteristicSpecification"
                },
                {
                    "name": "destinationNumber",
                    "description": "phone number of the called party",
                    "configurable": true,
```

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```
                    "valueType": "number",
                    "minCardinality": 1,
                    "maxCardinality": 1,
                    "isUnique": true,
                    "@type":"CharacteristicSpecification"
                },
                {
                    "name": "duration",
                    "description": "duration of the call",
                    "configurable": true,
                    "valueType": "number",
                    "minCardinality": 1,
                    "maxCardinality": 1,
                    "@type":"CharacteristicSpecification"
                },
                {
                    "name": "unit",
                    "description": "unit of the duration",
                    "configurable": true,
                    "valueType": "string",
                    "minCardinality": 0,
                    "maxCardinality": 1,
                    "characteristicValueSpecification": [
                        {
                            "isDefault": true,
                            "@type":"CharacteristicValueSpecification"
                        }
                    ],
                    "@type":"CharacteristicSpecification"
                },
                {
                    "name": "startDateTime",
                    "description": "startdate and starttime of the call",
                    "configurable": true,
                    "valueType": "dateTime",
                    "minCardinality": 0,
                    "maxCardinality": 1,
                    "@type":"CharacteristicSpecification"
                },
                {
                    "name": "endDateTime",
                    "description": "enddate and endtime of the call",
                    "configurable": true,
                    "valueType": "dateTime",
                    "minCardinality": 0,
                    "maxCardinality": 1,
                    "@type":"CharacteristicSpecification"
                }
            ],
            "validFor": {
                "startDateTime": "2022-02-20T17:00:00.000Z",
                "endDateTime": "2099-01-11T00:00:00.000Z"
            },
            "@type":"ServiceUsageSpecification"
        }
    },
    "reportingSystem": {
        "id": "504",
        "name": "APP-302",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
```


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```
        "id": "732",
        "name": "APP-641",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "ServiceUsageSpecificationCreateEvent"
}
```

Delete Event

Message example for ServiceUsageSpecificationDeleteEvent event

```
Content-Type: application/json

{
    "correlationId": "17b26b4d-cfe4",
    "description": "ServiceUsageSpecificationDeleteEvent illustration",
    "domain": "Commercial",
    "eventId": "4f3f-8fd6-67018c1f8b65",
    "eventTime": "2023-03-06T04:18:39.350Z",
    "eventType": "ServiceUsageSpecificationDeleteEvent",
    "priority": "2",
    "title": "ServiceUsageSpecificationDeleteEvent",
    "event": {
        "serviceUsageSpecification": {
            "id": "8b64-0cfd1a90c9ed",
            "href":
"https://api.csp.com/serviceUsageManagement/v5/serviceUsageSpecification/8b64-
0cfd1a90c9ed",
            "@type": "ServiceUsageSpecification"
        }
    },
    "reportingSystem": {
        "id": "504",
        "name": "APP-302",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "732",
        "name": "APP-641",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "ServiceUsageSpecificationDeleteEvent"
}
```

Attribute Value Change Event

Message example for ServiceUsageSpecificationAttributeValueChangeEvent event

```
Content-Type: application/json

{
    "correlationId": "e29100f8-79d2",
```

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```
    "description": "ServiceUsageSpecificationAttributeValueChangeEvent
illustration",
    "domain": "Commercial",
    "eventId": "4c10-a7ce-285f3d8d623f",
    "eventTime": "2023-03-06T04:18:39.349Z",
    "eventType": "ServiceUsageSpecificationAttributeValueChangeEvent",
    "priority": "5",
    "title": "ServiceUsageSpecificationAttributeValueChangeEvent",
    "event": {
        "serviceUsageSpecification": {
            "href":
"https://api.csp.com/serviceUsageManagement/v5/serviceUsageSpecification/8b64-
0cfd1a90c9ed",
            "id": "8b64-0cfd1a90c9ed",
            "validFor": {
                "startDateTime": "2022-02-20T17:00:00.000Z",
                "endDateTime": "2099-01-11T00:00:00.000Z"
            },
            "@type":"ServiceUsageSpecification"
        }
    },
    "reportingSystem": {
        "id": "504",
        "name": "APP-302",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "732",
        "name": "APP-641",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "ServiceUsageSpecificationAttributeValueChangeEvent"
}
```


State Change Event

Message example for ServiceUsageSpecificationStateChangeEvent event


```
Content-Type: application/json

{
    "correlationId": "e29100f8-79d2",
    "description": "ServiceUsageSpecificationAttributeValueChangeEvent
illustration",
    "domain": "Commercial",
    "eventId": "4c10-a7ce-285f3d8d623f",
    "eventTime": "2023-03-06T04:18:39.349Z",
    "eventType": "ServiceUsageSpecificationAttributeValueChangeEvent",
    "priority": "5",
    "title": "ServiceUsageSpecificationAttributeValueChangeEvent",
    "event": {
        "serviceUsageSpecification": {
            "href":
"https://api.csp.com/serviceUsageManagement/v5/serviceUsageSpecification/8b64-
0cfd1a90c9ed",
            "id": "8b64-0cfd1a90c9ed",
            "lifecycleStatus": "retired",
```


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```
            "@type":"ServiceUsageSpecification"
        }
    },
    "reportingSystem": {
        "id": "504",
        "name": "APP-302",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "732",
        "name": "APP-641",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "ServiceUsageSpecificationStateChangeEvent"
}
```


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API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

Operations on ServiceUsage

Retrieves a ServiceUsage by ID

`GET /serviceUsage/{id}?fields=…`

Description

This operation retrieves a ServiceUsage entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

A request for a previous Usage event. The criteria is the Usage id 5130d5131a23

Request

```
GET /serviceUsage/5130d5131a23
Content-Type: application/json
```

Response

```
200
```

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```
{
    "id": "5130d5131a23",
    "href":
"https://api.csp.com/serviceUsageManagement/v5/serviceUsage/5130d5131a23",
    "usageDate": "2020-11-20T16:07:00Z",
    "description": "Voice Call",
    "usageType": "Voice",
    "@type": "ServiceUsage",
    "service": {
        "id": "8132-b246a6746137",
        "href": "https://api.csp.com/serviceInventory/v5/service/8132-b246a6746137",
        "name": "Ivor voice service",
        "@type":"ServiceRef"
    },
    "usageSpecification": {
        "id": "8b64-0cfd1a90c9ed",
        "href":
"https://api.csp.com/serviceUsageManagement/v5/serviceUsageSpecification/8b64-
0cfd1a90c9ed",
        "name": "Voice Call Usage Specification",
        "@referredType": "ServiceUsageSpecification",
        "@type":"ServiceUsageSpecificationRef"
    },
    "relatedParty": [
        {
            "role": "Customer",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "id": "8185-b246a6746136",
              "href":
"https://serverlocation:port/partyManagement/v5/individual/8185-b246a6746136",
              "name": "Ivor Message",
              "@referredType": "Individual"
            }
        }
    ],
    "usageCharacteristic": [
        {
            "name": "originatingCountryCode",
            "value": "43",
            "valueType": "string",
            "@type":"StringCharacteristic"
        },
        {
            "name": "originatingNumber",
            "value": "676123456789",
            "valueType": "string",
            "@type": "StringCharacteristic"
        },
        {
            "name": "destinationCountryCode",
            "value": "49",
            "valueType": "string",
            "@type": "StringCharacteristic"
        },
        {
            "name": "destinationNumber",
            "value": "170123456789",
            "valueType": "string",
            "@type": "StringCharacteristic"
        },
        {
```


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```
            "name": "duration",
            "value": 20,
            "valueType": "number",
            "@type": "NumberCharacteristic"
        },
        {
            "name": "unit",
            "value": "SEC",
            "valueType": "string",
            "@type": "StringCharacteristic"
        }
    ]
}
```

List or find ServiceUsage objects

`GET /serviceUsage?fields=…`

Description

This operation list ServiceUsage entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieving previous Usage details for the usageType of 'Voice'.

Request

```
GET /serviceUsage?usageType=Voice
Content-Type: application/json
```

Response

```
200

[
    {
        "id": "5130d5131a23",
        "href":
"https://api.csp.com/serviceUsageManagement/v5/serviceUsage/5130d5131a23",
        "usageDate": "2020-11-20T16:07:00Z",
        "description": "Voice Call",
        "usageType": "Voice",
        "@type": "ServiceUsage",
        "service": {
            "id": "8132-b246a6746137",
            "href": "https://api.csp.com/serviceInventory/v5/service/8132-
b246a6746137",
            "name": "Ivor voice service",
            "@type":"ServiceRef"
        },
        "usageSpecification": {
            "id": "8b64-0cfd1a90c9ed",
```

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```
            "href":
"https://api.csp.com/serviceUsageManagement/v5/serviceUsageSpecification/8b64-
0cfd1a90c9ed",
            "name": "Voice Call Usage Specification",
            "@referredType": "ServiceUsageSpecification",
            "@type":"ServiceUsageSpecificationRef"
        },
        "relatedParty": [
            {
                "role": "Customer",
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "partyOrPartyRole": {
                  "@type": "PartyRef",
                  "id": "8185-b246a6746136",
                  "href":
"https://serverlocation:port/partyManagement/v5/individual/8185-b246a6746136",
                  "name": "Ivor Message",
                  "@referredType": "Individual"
                }
            }
        ],
        "usageCharacteristic": [
            {
                "name": "originatingCountryCode",
                "value": "43",
                "valueType": "string",
                "@type":"StringCharacteristic"
            },
            {
                "name": "originatingNumber",
                "value": "676123456789",
                "valueType": "string",
                "@type": "StringCharacteristic"
            },
            {
                "name": "destinationCountryCode",
                "value": "49",
                "valueType": "string",
                "@type": "StringCharacteristic"
            },
            {
                "name": "destinationNumber",
                "value": "170123456789",
                "valueType": "string",
                "@type": "StringCharacteristic"
            },
            {
                "name": "duration",
                "value": 20,
                "valueType": "number",
                "@type": "NumberCharacteristic"
            },
            {
                "name": "unit",
                "value": "SEC",
                "valueType": "string",
                "@type": "StringCharacteristic"
            }
        ]
    },
    {
        "id": "93c2-683ea281566c",
        "href": "https://api.csp.com/serviceUsageManagement/v5/serviceUsage/93c2-
```


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```
683ea281566c",
        "usageDate": "2020-11-20T08:13:16.000Z",
        "description": "Voice Call",
        "usageType": "Voice",
        "@type": "ServiceUsage",
        "service": {
            "id": "8132-b246a6746222",
            "href": "https://api.csp.com/serviceInventory/v5/service/8132-
b246a6746222",
            "name": "Ivor voice service",
            "@type":"ServiceRef"
        },
        "usageSpecification": {
            "id": "8b64-0cfd1a90c9ed",
            "href":
"https://api.csp.com/serviceUsageManagement/v5/serviceUsageSpecification/8b64-
0cfd1a90c9ed",
            "name": "Voice Call Usage Specification",
            "@referredType": "ServiceUsageSpecification",
            "@type":"ServiceUsageSpecificationRef"
        },
        "relatedParty": [
            {
                "role": "Customer",
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "partyOrPartyRole": {
                  "@type": "PartyRef",
                  "id": "8185-b246a6746136",
                  "href":
"https://serverlocation:port/partyManagement/v5/individual/8185-b246a6746136",
                  "name": "Ivor Message",
                  "@referredType": "Individual"
                }
            }
        ],
        "usageCharacteristic": [
            {
                "name": "originatingCountryCode",
                "value": "46",
                "valueType": "string",
                "@type":"StringCharacteristic"
            },
            {
                "name": "originatingNumber",
                "value": "676123459999",
                "valueType": "string",
                "@type": "StringCharacteristic"
            },
            {
                "name": "destinationCountryCode",
                "value": "43",
                "valueType": "string",
                "@type": "StringCharacteristic"
            },
            {
                "name": "destinationNumber",
                "value": "170123452222",
                "valueType": "string",
                "@type": "StringCharacteristic"
            },
            {
                "name": "duration",
                "value": 60,
```


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```
                "valueType": "number",
                "@type": "NumberCharacteristic"
            },
            {
                "name": "unit",
                "value": "SEC",
                "valueType": "string",
                "@type": "StringCharacteristic"
            }
        ]
    }
]
```

Here's an example of a request for retrieving multiple servicesUsage with field selector.

Request

```
GET /serviceUsage?fields=id,href,status,usageType,usageDate,@type
Content-Type: application/json
```

Response

```
200

[
    {
        "id": "93c2-683ea281566c",
        "href": "https://api.csp.com/serviceUsageManagement/v5/serviceUsage/93c2-
683ea281566c",
        "usageType": "Voice",
        "usageDate": "2020-11-20T08:13:16.000Z",
        "@type":"ServiceUsage"
    },
    {
        "id": "bbba-0c6e63171b78",
        "href": "https://api.csp.com/serviceUsageManagement/v5/serviceUsage/bbba-
0c6e63171b78",
        "usageType": "Voice",
        "usageDate": "2020-09-01T08:30:01.000Z",
        "@type":"ServiceUsage"
    }
]
```

Creates a ServiceUsage

`POST /serviceUsage?fields=…`

Description

This operation creates a ServiceUsage entity.

Mandatory Attributes

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Mandatory Attributes	Rule
service	
service.@type	
service.id	
usageCharacteristic	
usageCharacteristic.@type	
usageCharacteristic.name	
usageCharacteristic.value	
usageDate	
@type	

Usage samples

Here's an example of a request for creating ServiceUsage.

Request

```
POST /serviceUsage
Content-Type: application/json

{
    "description": "Voice Call",
    "usageType": "Voice",
    "@type": "ServiceUsage",
    "usageDate": "2033-08-25T12:19:14Z",
    "service": {
        "id": "8132-b246a6746137",
        "href": "http://serverlocation:port/serviceInventory/v5/service/8132-
b246a6746137",
        "name": "Ivor voice service",
        "@type":"ServiceRef"
    },
    "usageSpecification": {
        "id": "8b64-0cfd1a90c9ed",
        "href":
"http://serverlocation:port/serviceUsageManagement/v5/serviceUsageSpecification/8b64
-0cfd1a90c9ed",
        "name": "Voice Call Usage Specification",
        "@referredType": "ServiceUsageSpecification",
        "@type":"ServiceUsageSpecificationRef"
    },
    "relatedParty": [
        {
            "role": "Customer",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "id": "8185-b246a6746136",
              "href":
"http://serverlocation:port/partyManagement/v5/individual/8185-b246a6746136",
              "name": "Ivor Message",
              "@referredType": "Individual"
            }
```

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```
        }
    ],
    "usageCharacteristic": [
        {
            "name": "originatingCountryCode",
            "value": "43",
            "valueType": "string",
            "@type":"StringCharacteristic"
        },
        {
            "name": "originatingNumber",
            "value": "676123456789",
            "valueType": "string",
            "@type": "StringCharacteristic"
        },
        {
            "name": "destinationCountryCode",
            "value": "49",
            "valueType": "string",
            "@type": "StringCharacteristic"
        },
        {
            "name": "destinationNumber",
            "value": "170123456789",
            "valueType": "string",
            "@type": "StringCharacteristic"
        },
        {
            "name": "duration",
            "value": 20,
            "valueType": "number",
            "@type": "NumberCharacteristic"
        },
        {
            "name": "unit",
            "value": "SEC",
            "valueType": "string",
            "@type": "StringCharacteristic"
        }
    ]
}
```

Response

```
201

{
    "id": "5130d5131a23",
    "href":
"https://api.csp.com/serviceUsageManagement/v5/serviceUsage/5130d5131a23",
    "usageDate": "2020-11-20T16:07:00Z",
    "description": "Voice Call",
    "usageType": "Voice",
    "@type": "ServiceUsage",
    "service": {
        "id": "8132-b246a6746137",
        "href": "https://api.csp.com/serviceInventory/v5/service/8132-b246a6746137",
        "name": "Ivor voice service",
        "@type":"ServiceRef"
    },
```

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```
    "usageSpecification": {
        "id": "8b64-0cfd1a90c9ed",
        "href":
"https://api.csp.com/serviceUsageManagement/v5/serviceUsageSpecification/8b64-
0cfd1a90c9ed",
        "name": "Voice Call Usage Specification",
        "@referredType": "ServiceUsageSpecification",
        "@type":"ServiceUsageSpecificationRef"
    },
    "relatedParty": [
        {
            "role": "Customer",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "id": "8185-b246a6746136",
              "href":
"https://serverlocation:port/partyManagement/v5/individual/8185-b246a6746136",
              "name": "Ivor Message",
              "@referredType": "Individual"
            }
        }
    ],
    "usageCharacteristic": [
        {
            "name": "originatingCountryCode",
            "value": "43",
            "valueType": "string",
            "@type":"StringCharacteristic"
        },
        {
            "name": "originatingNumber",
            "value": "676123456789",
            "valueType": "string",
            "@type": "StringCharacteristic"
        },
        {
            "name": "destinationCountryCode",
            "value": "49",
            "valueType": "string",
            "@type": "StringCharacteristic"
        },
        {
            "name": "destinationNumber",
            "value": "170123456789",
            "valueType": "string",
            "@type": "StringCharacteristic"
        },
        {
            "name": "duration",
            "value": 20,
            "valueType": "number",
            "@type": "NumberCharacteristic"
        },
        {
            "name": "unit",
            "value": "SEC",
            "valueType": "string",
            "@type": "StringCharacteristic"
        }
    ]
}
```


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Updates partially a ServiceUsage

`PATCH /serviceUsage/{id}?fields=…`

Description

This operation allows partial updates of a ServiceUsage entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
usageDate	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
bundledServiceUsage	
description	
externalIdentifier	
href	
id	
isBundle	
relatedParty	
resourceUsage	
service	
usageCharacteristic	
usageSpecification	
usageType	

Usage samples

Here's an example of a request for partial update of ServiceUsage for application-json-patch-json PATCH type.

Request

```
PATCH /serviceUsage/5130d5131a23
Content-Type: application/json-patch+json
```

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```
[
    {
      "op": "replace",
      "path": "description",
      "value": "Voice Call service usage"
    }
  ]
```

Response

```
200

{
    "id": "5130d5131a23",
    "href":
"https://serverlocation:port/serviceUsagemanagement/v5/serviceUsage/5130d5131a23",
    "usageDate": "2020-11-20T16:07:00Z",
    "description": "Voice Call service usage",
    "usageType": "Voice",
    "@type": "ServiceUsage",
    "service": {
        "id": "8132-b246a6746137",
        "href": "https://serverlocation:port/serviceInventory/v5/service/8132-
b246a6746137",
        "name": "Ivor voice service",
        "@type":"Service"
    },
    "usageSpecification": {
        "id": "8b64-0cfd1a90c9ed",
        "href":
"https://serverlocation:port/serviceUsageManagement/v5/serviceUsageSpecification/8b6
4-0cfd1a90c9ed",
        "name": "Voice Call Usage Specification",
        "@referredType": "ServiceUsageSpecification",
        "@type":"ServiceUsageSpecificationRef"
    },
    "relatedParty": [
        {
            "role": "Customer",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "id": "8185-b246a6746136",
              "href":
"https://serverlocation:port/partyManagement/v5/individual/8185-b246a6746136",
              "name": "Ivor Message",
              "@referredType": "Individual"
            }
        }
    ],
    "usageCharacteristic": [
        {
            "name": "originatingCountryCode",
            "value": "90",
            "valueType": "string",
            "@type":"StringCharacteristic"
        },
        {
            "name": "originatingNumber",
```

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```
            "value": "676123456789",
            "valueType": "string",
            "@type": "StringCharacteristic"
        },
        {
            "name": "destinationCountryCode",
            "value": "49",
            "valueType": "string",
            "@type": "StringCharacteristic"
        },
        {
            "name": "destinationNumber",
            "value": "170123456789",
            "valueType": "string",
            "@type": "StringCharacteristic"
        },
        {
            "name": "duration",
            "value": 20,
            "valueType": "number",
            "@type": "NumberCharacteristic"
        },
        {
            "name": "unit",
            "value": "SEC",
            "valueType": "string",
            "@type": "StringCharacteristic"
        }
    ]
}
```

Here's an example of a request for partial update of ServiceUsage for application-merge-patch-json PATCH type.

Request

```
PATCH /serviceUsage/5130d5131a23
Content-Type: application/merge-patch+json

{
    "@type": "ServiceUsage",
    "usageCharacteristic": [
        {
            "@type":"StringCharacteristic",
            "name": "originatingCountryCode",
            "value": "90",
            "valueType": "string"

        },
        {
            "name": "originatingNumber",
            "value": "676123456789",
            "valueType": "string",
            "@type": "StringCharacteristic"
        },
        {
            "name": "destinationCountryCode",
            "value": "49",
            "valueType": "string",
            "@type": "StringCharacteristic"
```

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```
        },
        {
            "name": "destinationNumber",
            "value": "170123456789",
            "valueType": "string",
            "@type": "StringCharacteristic"
        },
        {
            "name": "duration",
            "value": 20,
            "valueType": "number",
            "@type": "NumberCharacteristic"
        },
        {
            "name": "unit",
            "value": "SEC",
            "valueType": "string",
            "@type": "StringCharacteristic"
        },
        {
            "name": "Call quality",
            "value": "Excellent",
            "valueType": "string",
            "@type": "StringCharacteristic"
        }
    ]
}
```

Response

```
200

{
    "id": "5130d5131a23",
    "href":
"https://serverlocation:port/serviceUsagemanagement/v5/serviceUsage/5130d5131a23",
    "usageDate": "2020-11-20T16:07:00Z",
    "description": "Voice Call",
    "usageType": "Voice",
    "@type": "ServiceUsage",
    "service": {
        "id": "8132-b246a6746137",
        "href": "https://serverlocation:port/serviceInventory/v5/service/8132-
b246a6746137",
        "name": "Ivor voice service",
        "@type":"ServiceRef"
    },
    "usageSpecification": {
        "id": "8b64-0cfd1a90c9ed",
        "href":
"https://serverlocation:port/serviceUsageManagement/v5/serviceUsageSpecification/8b6
4-0cfd1a90c9ed",
        "name": "Voice Call Usage Specification",
        "@referredType": "ServiceUsageSpecification",
        "@type":"ServiceUsageSpecificationRef"
    },
    "relatedParty": [
        {
            "role": "Customer",
            "@type": "RelatedPartyRefOrPartyRoleRef",
```

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```
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "id": "8185-b246a6746136",
              "href":
"https://serverlocation:port/partyManagement/v5/individual/8185-b246a6746136",
              "name": "Ivor Message",
              "@referredType": "Individual"
            }
        }
    ],
    "usageCharacteristic": [
        {
            "name": "originatingCountryCode",
            "value": "90",
            "valueType": "string",
            "@type":"StringCharacteristic"
        },
        {
            "name": "originatingNumber",
            "value": "676123456789",
            "valueType": "string",
            "@type": "StringCharacteristic"
        },
        {
            "name": "destinationCountryCode",
            "value": "49",
            "valueType": "string",
            "@type": "StringCharacteristic"
        },
        {
            "name": "destinationNumber",
            "value": "170123456789",
            "valueType": "string",
            "@type": "StringCharacteristic"
        },
        {
            "name": "duration",
            "value": 20,
            "valueType": "number",
            "@type": "NumberCharacteristic"
        },
        {
            "name": "unit",
            "value": "SEC",
            "valueType": "string",
            "@type": "StringCharacteristic"
        },
        {
            "name": "Call quality",
            "value": "Excellent",
            "valueType": "string",
            "@type": "StringCharacteristic"
        }
    ]
}
```


Here's an example of a request for partial update of ServiceUsage for application-json-patch-query-json PATCH type.

Request

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```
PATCH /serviceUsage/5130d5131a23
Content-Type: application/json-patch-query+json

[
    {
      "op": "replace",
      "path": "/usageCharacteristic/value?usageCharacteristic.name=duration",
      "value": 60
    }
]
```

Response

```
200

{
    "id": "5130d5131a23",
    "href":
"https://serverlocation:port/serviceUsagemanagement/v5/serviceUsage/5130d5131a23",
    "usageDate": "2020-11-20T16:07:00Z",
    "description": "Voice Call",
    "usageType": "Voice",
    "@type": "ServiceUsage",
    "service": {
        "id": "8132-b246a6746137",
        "href": "https://serverlocation:port/serviceInventory/v5/service/8132-
b246a6746137",
        "name": "Ivor voice service",
        "@type":"ServiceRef"
    },
    "usageSpecification": {
        "id": "8b64-0cfd1a90c9ed",
        "href":
"https://serverlocation:port/serviceUsageManagement/v5/serviceUsageSpecification/8b6
4-0cfd1a90c9ed",
        "name": "Voice Call Usage Specification",
        "@referredType": "ServiceUsageSpecification",
        "@type":"ServiceUsageSpecificationRef"
    },
    "relatedParty": [
        {
            "role": "Customer",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "id": "8185-b246a6746136",
              "href":
"https://serverlocation:port/partyManagement/v5/individual/8185-b246a6746136",
              "name": "Ivor Message",
              "@referredType": "Individual"
            }
        }
    ],
    "usageCharacteristic": [
        {
            "name": "originatingCountryCode",
            "value": "90",
            "valueType": "string",
            "@type":"StringCharacteristic"
```

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```
        },
        {
            "name": "originatingNumber",
            "value": "676123456789",
            "valueType": "string",
            "@type": "StringCharacteristic"
        },
        {
            "name": "destinationCountryCode",
            "value": "49",
            "valueType": "string",
            "@type": "StringCharacteristic"
        },
        {
            "name": "destinationNumber",
            "value": "170123456789",
            "valueType": "string",
            "@type": "StringCharacteristic"
        },
        {
            "name": "duration",
            "value": 60,
            "valueType": "number",
            "@type": "NumberCharacteristic"
        },
        {
            "name": "unit",
            "value": "SEC",
            "valueType": "string",
            "@type": "StringCharacteristic"
        }
    ]
}
```

Here's an example of a request for partial update of ServiceUsage for application-json PATCH type.

Request

```
PATCH /serviceUsage/5130d5131a23
Content-Type: application/json

{
    "@type": "ServiceUsage",
    "usageCharacteristic": [
        {
            "@type":"StringCharacteristic",
            "name": "originatingCountryCode",
            "value": "90",
            "valueType": "string"

        },
        {
            "name": "originatingNumber",
            "value": "676123456789",
            "valueType": "string",
            "@type": "StringCharacteristic"
        },
        {
            "name": "destinationCountryCode",
```

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```
            "value": "49",
            "valueType": "string",
            "@type": "StringCharacteristic"
        },
        {
            "name": "destinationNumber",
            "value": "170123456789",
            "valueType": "string",
            "@type": "StringCharacteristic"
        },
        {
            "name": "duration",
            "value": 20,
            "valueType": "number",
            "@type": "NumberCharacteristic"
        },
        {
            "name": "unit",
            "value": "SEC",
            "valueType": "string",
            "@type": "StringCharacteristic"
        },
        {
            "name": "Call quality",
            "value": "Excellent",
            "valueType": "string",
            "@type": "StringCharacteristic"
        }
    ]
}
```

Response

```
200

{
    "id": "5130d5131a23",
    "href":
"https://serverlocation:port/serviceUsagemanagement/v5/serviceUsage/5130d5131a23",
    "usageDate": "2020-11-20T16:07:00Z",
    "description": "Voice Call",
    "usageType": "Voice",
    "@type": "ServiceUsage",
    "service": {
        "id": "8132-b246a6746137",
        "href": "https://serverlocation:port/serviceInventory/v5/service/8132-
b246a6746137",
        "name": "Ivor voice service",
        "@type":"ServiceRef"
    },
    "usageSpecification": {
        "id": "8b64-0cfd1a90c9ed",
        "href":
"https://serverlocation:port/serviceUsageManagement/v5/serviceUsageSpecification/8b6
4-0cfd1a90c9ed",
        "name": "Voice Call Usage Specification",
        "@referredType": "ServiceUsageSpecification",
        "@type":"ServiceUsageSpecificationRef"
    },
    "relatedParty": [
```

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```
        {
            "role": "Customer",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "id": "8185-b246a6746136",
              "href":
"https://serverlocation:port/partyManagement/v5/individual/8185-b246a6746136",
              "name": "Ivor Message",
              "@referredType": "Individual"
            }
        }
    ],
    "usageCharacteristic": [
        {
            "name": "originatingCountryCode",
            "value": "90",
            "valueType": "string",
            "@type":"StringCharacteristic"
        },
        {
            "name": "originatingNumber",
            "value": "676123456789",
            "valueType": "string",
            "@type": "StringCharacteristic"
        },
        {
            "name": "destinationCountryCode",
            "value": "49",
            "valueType": "string",
            "@type": "StringCharacteristic"
        },
        {
            "name": "destinationNumber",
            "value": "170123456789",
            "valueType": "string",
            "@type": "StringCharacteristic"
        },
        {
            "name": "duration",
            "value": 20,
            "valueType": "number",
            "@type": "NumberCharacteristic"
        },
        {
            "name": "unit",
            "value": "SEC",
            "valueType": "string",
            "@type": "StringCharacteristic"
        },
        {
            "name": "Call quality",
            "value": "Excellent",
            "valueType": "string",
            "@type": "StringCharacteristic"
        }
    ]
}
```


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Deletes a ServiceUsage

DELETE /serviceUsage/{id}

Description

This operation deletes a ServiceUsage entity.

Usage samples

Here's an example of a request for deleting a specific ServiceUsage.

Request

```
DELETE /serviceUsage/5130d5131a23
Content-Type: application/json
```

Response

```
204

{
    "id": "5130d5131a23",
    "href":
"https://api.csp.com/serviceUsageManagement/v5/serviceUsage/5130d5131a23",
    "@type": "ServiceUsage"
}
```

Operations on ServiceUsageSpecification

List or find ServiceUsageSpecification objects

GET /serviceUsageSpecification?fields=…

Description

This operation list ServiceUsageSpecification entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieving previous ServiceUsageSpecifications which are at version 2.5

Request

```
GET /serviceUsageSpecification?version=2.5
Content-Type: application/json
```

Response

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```
200

[
    {
        "id": "8b64-0cfd1a90c9ed",
        "href":
"https://api.csp.com/serviceUsageManagement/v5/serviceUsageSpecification/8b64-
0cfd1a90c9ed",
        "description": "Specification of usage for voice calls",
        "isBundle": false,
        "lastUpdate": "2020-07-07T08:34:13.831Z",
        "lifecycleStatus": "active",
        "name": "Voice-Usage-Specification",
        "version": "2.5",
        "relatedParty": [
            {
                "role": "Customer",
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "partyOrPartyRole": {
                  "@type": "PartyRef",
                  "id": "8185-b246a6746136",
                  "href":
"http://serverlocation:port/partyManagement/v5/individual/8185-b246a6746136",
                  "name": "Ivor Message",
                  "@referredType": "Individual"
                }
            }
        ],
        "specCharacteristic": [
            {
                "name": "Originating Country Code",
                "description": "Country Code of the caller",
                "configurable": true,
                "valueType": "string",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type":"CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            },
            {
                "name": "Originating Number",
                "description": "Phone number of the caller",
                "configurable": true,
                "valueType": "string",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type":"CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            },
            {
                "name": "Destination Country Code",
                "description": "Country Code of the recipient",
                "configurable": true,
                "valueType": "string",
```


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```
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type":"CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            },
            {
                "name": "Destination Number",
                "description": "Phone number of the recipient",
                "configurable": true,
                "valueType": "string",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type":"CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            },
            {
                "name": "Duration",
                "description": "Duration of the call in seconds",
                "configurable": true,
                "valueType": "number",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type":"CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            },
            {
                "name": "startDateTime",
                "description": "Start date and start time of the call",
                "configurable": true,
                "valueType": "dateTime",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type":"CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            },
            {
                "name": "endDateTime",
                "description": "End date and end time of the call",
                "configurable": true,
                "valueType": "dateTime",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type":"CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
```


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```
                }
            }
        ],
        "@type":"ServiceUsageSpecification"
    }
]
```

Retrieves a ServiceUsageSpecification by ID

`GET /serviceUsageSpecification/{id}?fields=…`

Description

This operation retrieves a ServiceUsageSpecification entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

A request for a specific ServiceUsageSpecification with an id 8b64-0cfd1a90c9ed

Request

```
GET /serviceUsageSpecification/8b64-0cfd1a90c9ed
Content-Type: application/json
```

Response

```
200

{
    "id": "8b64-0cfd1a90c9ed",
    "href":
"https://api.csp.com/serviceUsageManagement/v5/serviceUsageSpecification/8b64-
0cfd1a90c9ed",
    "description": "Specification of usage for voice calls",
    "isBundle": false,
    "lastUpdate": "2020-07-07T08:34:13.831Z",
    "lifecycleStatus": "active",
    "name": "Voice-Usage-Specification",
    "version": "2.5",
    "relatedParty": [
        {
            "role": "Customer",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "id": "8185-b246a6746136",
              "href":
"http://serverlocation:port/partyManagement/v5/individual/8185-b246a6746136",
              "name": "Ivor Message",
              "@referredType": "Individual"
            }
        }
    ],
```

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```
    "specCharacteristic": [
        {
            "name": "Originating Country Code",
            "description": "Country Code of the caller",
            "configurable": true,
            "valueType": "string",
            "minCardinality": 1,
            "maxCardinality": 1,
            "@type":"CharacteristicSpecification",
            "extensible": true,
            "isUnique": true,
            "validFor": {
                "startDateTime": "2020-09-23T16:42:23Z"
            }

        },
        {
            "name": "Originating Number",
            "description": "Phone number of the caller",
            "configurable": true,
            "valueType": "string",
            "minCardinality": 1,
            "maxCardinality": 1,
            "@type":"CharacteristicSpecification",
            "extensible": true,
            "isUnique": true,
            "validFor": {
                "startDateTime": "2020-09-23T16:42:23Z"
            }
        },
        {
            "name": "Destination Country Code",
            "description": "Country Code of the recipient",
            "configurable": true,
            "valueType": "string",
            "minCardinality": 1,
            "maxCardinality": 1,
            "@type":"CharacteristicSpecification",
            "extensible": true,
            "isUnique": true,
            "validFor": {
                "startDateTime": "2020-09-23T16:42:23Z"
            }
        },
        {
            "name": "Destination Number",
            "description": "Phone number of the recipient",
            "configurable": true,
            "valueType": "string",
            "minCardinality": 1,
            "maxCardinality": 1,
            "@type":"CharacteristicSpecification",
            "extensible": true,
            "isUnique": true,
            "validFor": {
                "startDateTime": "2020-09-23T16:42:23Z"
            }
        },
        {
            "name": "Duration",
            "description": "Duration of the call in seconds",
            "configurable": true,
            "valueType": "number",
```


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```
            "minCardinality": 1,
            "maxCardinality": 1,
            "@type":"CharacteristicSpecification",
            "extensible": true,
            "isUnique": true,
            "validFor": {
                "startDateTime": "2020-09-23T16:42:23Z"
            }
        },
        {
            "name": "startDateTime",
            "description": "Start date and start time of the call",
            "configurable": true,
            "valueType": "dateTime",
            "minCardinality": 1,
            "maxCardinality": 1,
            "@type":"CharacteristicSpecification",
            "extensible": true,
            "isUnique": true,
            "validFor": {
                "startDateTime": "2020-09-23T16:42:23Z"
            }
        },
        {
            "name": "endDateTime",
            "description": "End date and end time of the call",
            "configurable": true,
            "valueType": "dateTime",
            "minCardinality": 1,
            "maxCardinality": 1,
            "@type":"CharacteristicSpecification",
            "extensible": true,
            "isUnique": true,
            "validFor": {
                "startDateTime": "2020-09-23T16:42:23Z"
            }
        }
    ],
    "@type":"ServiceUsageSpecification"
}
```


Creates a ServiceUsageSpecification

`POST /serviceUsageSpecification?fields=…`

Description

This operation creates a ServiceUsageSpecification entity.

Mandatory Attributes

Mandatory Attributes	Rule
lifecycleStatus	
name	
@type	

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Usage samples

Here's an example of a request for creating a service usage specification

Request

```
POST /serviceUsageSpecification
Content-Type: application/json

{
    "description": "Specification of usage for voice calls",
    "lifecycleStatus": "string",
    "name": "Voice-Usage-Specification",
    "version": "2.5",
    "isBundle": true,
    "bundledServiceUsageSpecification" : [
        {   "id": "90c9ed-8b64",
            "@type":"ServiceUsageSpecificationRef",
            "name": "Local calls"
        },
        {   "id": "435d-87884",
            "@type":"ServiceUsageSpecificationRef",
            "name": "Long-dinstance calls"
        }
    ],
    "relatedParty": [
        {
            "role": "Customer",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "id": "8185-b246a6746136",
              "href":
"http://serverlocation:port/partyManagement/v5/individual/8185-b246a6746136",
              "name": "Ivor Message",
              "@referredType": "Individual"
            }
        }
    ],
    "specCharacteristic": [
        {
            "name": "Originating Country Code",
            "description": "Country Code of the caller",
            "configurable": true,
            "valueType": "string",
            "minCardinality": 1,
            "maxCardinality": 1,
            "@type":"CharacteristicSpecification",
            "extensible": true,
            "isUnique": true,
            "validFor": {
                "startDateTime": "2020-09-23T16:42:23Z"
            }
        },
        {
            "name": "Originating Number",
            "description": "Phone number of the caller",
            "configurable": true,
            "valueType": "string",
            "minCardinality": 1,
            "maxCardinality": 1,
```

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```
            "@type":"CharacteristicSpecification",
            "extensible": true,
            "isUnique": true,
            "validFor": {
                "startDateTime": "2020-09-23T16:42:23Z"
            }
        },
        {
            "name": "Destination Country Code",
            "description": "Country Code of the recipient",
            "configurable": true,
            "valueType": "string",
            "minCardinality": 1,
            "maxCardinality": 1,
            "@type":"CharacteristicSpecification",
            "extensible": true,
            "isUnique": true,
            "validFor": {
                "startDateTime": "2020-09-23T16:42:23Z"
            }
        },
        {
            "name": "Destination Number",
            "description": "Phone number of the recipient",
            "configurable": true,
            "valueType": "string",
            "minCardinality": 1,
            "maxCardinality": 1,
            "@type":"CharacteristicSpecification",
            "extensible": true,
            "isUnique": true,
            "validFor": {
                "startDateTime": "2020-09-23T16:42:23Z"
            }
        },
        {
            "name": "Duration",
            "description": "Duration of the call in seconds",
            "configurable": true,
            "valueType": "number",
            "minCardinality": 1,
            "maxCardinality": 1,
            "@type":"CharacteristicSpecification",
            "extensible": true,
            "isUnique": true,
            "validFor": {
                "startDateTime": "2020-09-23T16:42:23Z"
            }
        },
        {
            "name": "startDateTime",
            "description": "Start date and start time of the call",
            "configurable": true,
            "valueType": "dateTime",
            "minCardinality": 1,
            "maxCardinality": 1,
            "@type":"CharacteristicSpecification",
            "extensible": true,
            "isUnique": true,
            "validFor": {
                "startDateTime": "2020-09-23T16:42:23Z"
            }
        },
```


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```
        {
            "name": "endDateTime",
            "description": "End date and end time of the call",
            "configurable": true,
            "valueType": "dateTime",
            "minCardinality": 1,
            "maxCardinality": 1,
            "@type":"CharacteristicSpecification",
            "extensible": true,
            "isUnique": true,
            "validFor": {
                "startDateTime": "2020-09-23T16:42:23Z"
            }
        }
    ],
    "@type":"ServiceUsageSpecification"
}
```

Response

```
201

{
    "id": "8b64-0cfd1a90c9ed",
    "href":
"https://api.csp.com/serviceUsageManagement/v5/serviceUsageSpecification/8b64-
0cfd1a90c9ed",
    "description": "Specification of usage for voice calls",
    "isBundle": false,
    "lastUpdate": "2020-07-07T08:34:13.831Z",
    "lifecycleStatus": "active",
    "name": "Voice-Usage-Specification",
    "version": "2.5",
    "relatedParty": [
        {
            "role": "Customer",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "id": "8185-b246a6746136",
              "href":
"http://serverlocation:port/partyManagement/v5/individual/8185-b246a6746136",
              "name": "Ivor Message",
              "@referredType": "Individual"
            }
        }
    ],
    "specCharacteristic": [
        {
            "name": "Originating Country Code",
            "description": "Country Code of the caller",
            "configurable": true,
            "valueType": "string",
            "minCardinality": 1,
            "maxCardinality": 1,
            "@type":"CharacteristicSpecification",
            "extensible": true,
            "isUnique": true,
            "validFor": {
                "startDateTime": "2020-09-23T16:42:23Z"
```

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```
            }

        },
        {
            "name": "Originating Number",
            "description": "Phone number of the caller",
            "configurable": true,
            "valueType": "string",
            "minCardinality": 1,
            "maxCardinality": 1,
            "@type":"CharacteristicSpecification",
            "extensible": true,
            "isUnique": true,
            "validFor": {
                "startDateTime": "2020-09-23T16:42:23Z"
            }
        },
        {
            "name": "Destination Country Code",
            "description": "Country Code of the recipient",
            "configurable": true,
            "valueType": "string",
            "minCardinality": 1,
            "maxCardinality": 1,
            "@type":"CharacteristicSpecification",
            "extensible": true,
            "isUnique": true,
            "validFor": {
                "startDateTime": "2020-09-23T16:42:23Z"
            }
        },
        {
            "name": "Destination Number",
            "description": "Phone number of the recipient",
            "configurable": true,
            "valueType": "string",
            "minCardinality": 1,
            "maxCardinality": 1,
            "@type":"CharacteristicSpecification",
            "extensible": true,
            "isUnique": true,
            "validFor": {
                "startDateTime": "2020-09-23T16:42:23Z"
            }
        },
        {
            "name": "Duration",
            "description": "Duration of the call in seconds",
            "configurable": true,
            "valueType": "number",
            "minCardinality": 1,
            "maxCardinality": 1,
            "@type":"CharacteristicSpecification",
            "extensible": true,
            "isUnique": true,
            "validFor": {
                "startDateTime": "2020-09-23T16:42:23Z"
            }
        },
        {
            "name": "startDateTime",
            "description": "Start date and start time of the call",
            "configurable": true,
```


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```
            "valueType": "dateTime",
            "minCardinality": 1,
            "maxCardinality": 1,
            "@type":"CharacteristicSpecification",
            "extensible": true,
            "isUnique": true,
            "validFor": {
                "startDateTime": "2020-09-23T16:42:23Z"
            }
        },
        {
            "name": "endDateTime",
            "description": "End date and end time of the call",
            "configurable": true,
            "valueType": "dateTime",
            "minCardinality": 1,
            "maxCardinality": 1,
            "@type":"CharacteristicSpecification",
            "extensible": true,
            "isUnique": true,
            "validFor": {
                "startDateTime": "2020-09-23T16:42:23Z"
            }
        }
    ],
    "@type":"ServiceUsageSpecification"
}
```


Updates partially a ServiceUsageSpecification

`PATCH /serviceUsageSpecification/{id}?fields=…`

Description

This operation allows partial updates of a ServiceUsageSpecification entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
href	
id	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
attachment	

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Patchable Attributes	Rule
bundledServiceUsageSpecification	
constraint	
description	
entitySpecRelationship	
isBundle	
lastUpdate	
lifecycleStatus	
name	
relatedParty	
resourceUsageSpecification	
serviceSpecification	
serviceUsageSpecRelationship	
specCharacteristic	
targetEntitySchema	
validFor	
version	

Usage samples

Here's an example of a request for creating a service usage specification

Request

```
PATCH /serviceUsageSpecification/{id}
Content-Type: application/json-patch+json

[
    {
      "op": "replace",
      "path": "lifecycleStatus",
      "value": "retired"
    }
  ]
```

Response

```
200

[
    {
        "id": "8b64-0cfd1a90c9ed",
        "href":
"https://api.csp.com/serviceUsageManagement/v5/serviceUsageSpecification/8b64-
```

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```
0cfd1a90c9ed",
        "description": "Specification of usage for voice calls",
        "isBundle": false,
        "lastUpdate": "2020-07-07T08:34:13.831Z",
        "lifecycleStatus": "retired",
        "name": "Voice-Usage-Specification",
        "version": "2.5",
        "relatedParty": [
            {
                "role": "Customer",
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "partyOrPartyRole": {
                  "@type": "PartyRef",
                  "id": "8185-b246a6746136",
                  "href":
"http://serverlocation:port/partyManagement/v5/individual/8185-b246a6746136",
                  "name": "Ivor Message",
                  "@referredType": "Individual"
                }
            }
        ],
        "specCharacteristic": [
            {
                "name": "Originating Country Code",
                "description": "Country Code of the caller",
                "configurable": true,
                "valueType": "string",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type":"CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            },
            {
                "name": "Originating Number",
                "description": "Phone number of the caller",
                "configurable": true,
                "valueType": "string",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type":"CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            },
            {
                "name": "Destination Country Code",
                "description": "Country Code of the recipient",
                "configurable": true,
                "valueType": "string",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type":"CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
```


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```
            },
            {
                "name": "Destination Number",
                "description": "Phone number of the recipient",
                "configurable": true,
                "valueType": "string",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type":"CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            },
            {
                "name": "Duration",
                "description": "Duration of the call in seconds",
                "configurable": true,
                "valueType": "number",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type":"CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            },
            {
                "name": "startDateTime",
                "description": "Start date and start time of the call",
                "configurable": true,
                "valueType": "dateTime",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type":"CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            },
            {
                "name": "endDateTime",
                "description": "End date and end time of the call",
                "configurable": true,
                "valueType": "dateTime",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type":"CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            }
        ],
        "@type":"ServiceUsageSpecification"
    }
]
```


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Here's an example of a request for creating a service usage specification

Request

```
PATCH /serviceUsageSpecification/{id}
Content-Type: application/merge-patch+json

    {
        "description": "Specification of usage for voice calls",
        "isBundle": false,
        "lastUpdate": "2020-07-07T08:34:13.831Z",
        "lifecycleStatus": "retired",
        "name": "Voice-Usage-Specification",
        "version": "2.5",
        "serviceUsageSpecRelationship": [
            {
                "id": "8b64-0cfd1a91r0xx",
                "href":
"https://api.csp.com/serviceUsageManagement/v5/serviceUsageSpecification/8b64-
0cfd1a91r0xx",
                "name": "voip-Usage-Specification",
                "relationshipType": "dependency",
                "@type":"ServiceUsageSpecRelationship"
            }
        ],
        "@type":"ServiceUsageSpecification"
    }
```

Response

```
200

    {
        "id": "8b64-0cfd1a90c9ed",
        "href":
"https://api.csp.com/serviceUsageManagement/v5/serviceUsageSpecification/8b64-
0cfd1a90c9ed",
        "description": "Specification of usage for voice calls",
        "isBundle": false,
        "lastUpdate": "2020-07-07T08:34:13.831Z",
        "lifecycleStatus": "active",
        "name": "Voice-Usage-Specification",
        "version": "2.5",
        "relatedParty": [
            {
                "role": "Customer",
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "partyOrPartyRole": {
                    "@type": "PartyRef",
                    "id": "8185-b246a6746136",
                    "href":
"http://serverlocation:port/partyManagement/v5/individual/8185-b246a6746136",
                    "name": "Ivor Message",
                    "@referredType": "Individual"
                }
            }
        ],
        "specCharacteristic": [
            {
```

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```
                "name": "Originating Country Code",
                "description": "Country Code of the caller",
                "configurable": true,
                "valueType": "string",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type": "CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            },
            {
                "name": "Originating Number",
                "description": "Phone number of the caller",
                "configurable": true,
                "valueType": "string",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type": "CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            },
            {
                "name": "Destination Country Code",
                "description": "Country Code of the recipient",
                "configurable": true,
                "valueType": "string",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type": "CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            },
            {
                "name": "Destination Number",
                "description": "Phone number of the recipient",
                "configurable": true,
                "valueType": "string",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type": "CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            },
            {
                "name": "Duration",
                "description": "Duration of the call in seconds",
                "configurable": true,
                "valueType": "number",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type": "CharacteristicSpecification",
```


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```
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            },
            {
                "name": "startDateTime",
                "description": "Start date and start time of the call",
                "configurable": true,
                "valueType": "dateTime",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type": "CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            },
            {
                "name": "endDateTime",
                "description": "End date and end time of the call",
                "configurable": true,
                "valueType": "dateTime",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type": "CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            }
        ],
        "@type": "ServiceUsageSpecification",
        "serviceUsageSpecRelationship": [
            {
                "id": "8b64-0cfd1a91r0xx",
                "href":
"https://api.csp.com/serviceUsageManagement/v5/serviceUsageSpecification/8b64-
0cfd1a91r0xx",
                "name": "voip-Usage-Specification",
                "@type": "ServiceUsageSpecRelationship"
            }
        ]
    }
```

Here's an example of a request for creating a service usage specification

Request

```
PATCH /serviceUsageSpecification/{id}
Content-Type: application/json-patch-query+json

[
    {
      "op": "replace",
      "path": "/specCharacteristic/value?specCharacteristic.name='Originating
Country Code'",
```

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```
      "value": "ISO Country Code of the caller"
    }
]
```

Response

```
200

[
    {
        "id": "8b64-0cfd1a90c9ed",
        "href":
"https://api.csp.com/serviceUsageManagement/v5/serviceUsageSpecification/8b64-
0cfd1a90c9ed",
        "description": "Specification of usage for voice calls",
        "isBundle": false,
        "lastUpdate": "2020-07-07T08:34:13.831Z",
        "lifecycleStatus": "active",
        "name": "Voice-Usage-Specification",
        "version": "2.5",
        "relatedParty": [
            {
                "role": "Customer",
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "partyOrPartyRole": {
                  "@type": "PartyRef",
                  "id": "8185-b246a6746136",
                  "href":
"http://serverlocation:port/partyManagement/v5/individual/8185-b246a6746136",
                  "name": "Ivor Message",
                  "@referredType": "Individual"
                }
            }
        ],
        "specCharacteristic": [
            {
                "name": "Originating Country Code",
                "description": "ISO Country Code of the caller",
                "configurable": true,
                "valueType": "string",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type":"CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            },
            {
                "name": "Originating Number",
                "description": "Phone number of the caller",
                "configurable": true,
                "valueType": "string",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type":"CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
```

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```
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            },
            {
                "name": "Destination Country Code",
                "description": "Country Code of the recipient",
                "configurable": true,
                "valueType": "string",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type":"CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            },
            {
                "name": "Destination Number",
                "description": "Phone number of the recipient",
                "configurable": true,
                "valueType": "string",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type":"CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            },
            {
                "name": "Duration",
                "description": "Duration of the call in seconds",
                "configurable": true,
                "valueType": "number",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type":"CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            },
            {
                "name": "startDateTime",
                "description": "Start date and start time of the call",
                "configurable": true,
                "valueType": "dateTime",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type":"CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            },
            {
                "name": "endDateTime",
                "description": "End date and end time of the call",
                "configurable": true,
```


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```
                "valueType": "dateTime",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type":"CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            }
        ],
        "@type":"ServiceUsageSpecification",
        "serviceUsageSpecRelationship": [
            {
                "id": "8b64-0cfd1a91r0xx",
                "href":
"https://api.csp.com/serviceUsageManagement/v5/serviceUsageSpecification/8b64-
0cfd1a91r0xx",
                "name": "voip-Usage-Specification",
                "@type":"ServiceUsageSpecRelationship"
            }
        ]
    }
]
```

Here's an example of a request for creating a service usage specification

Request

```
PATCH /serviceUsageSpecification/{id}
Content-Type: application/json

    {
        "description": "Specification of usage for voice calls",
        "isBundle": false,
        "lastUpdate": "2020-07-07T08:34:13.831Z",
        "lifecycleStatus": "retired",
        "name": "Voice-Usage-Specification",
        "version": "2.5",
        "serviceUsageSpecRelationship": [
            {
                "id": "8b64-0cfd1a91r0xx",
                "href":
"https://api.csp.com/serviceUsageManagement/v5/serviceUsageSpecification/8b64-
0cfd1a91r0xx",
                "name": "voip-Usage-Specification",
                "relationshipType": "dependency",
                "@type":"ServiceUsageSpecRelationship"
            }
        ],
        "@type":"ServiceUsageSpecification"
    }
```

Response

```
200
```

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```
    {
        "id": "8b64-0cfd1a90c9ed",
        "href":
"https://api.csp.com/serviceUsageManagement/v5/serviceUsageSpecification/8b64-
0cfd1a90c9ed",
        "description": "Specification of usage for voice calls",
        "isBundle": false,
        "lastUpdate": "2020-07-07T08:34:13.831Z",
        "lifecycleStatus": "active",
        "name": "Voice-Usage-Specification",
        "version": "2.5",
        "relatedParty": [
            {
                "role": "Customer",
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "partyOrPartyRole": {
                    "@type": "PartyRef",
                    "id": "8185-b246a6746136",
                    "href":
"http://serverlocation:port/partyManagement/v5/individual/8185-b246a6746136",
                    "name": "Ivor Message",
                    "@referredType": "Individual"
                }
            }
        ],
        "specCharacteristic": [
            {
                "name": "Originating Country Code",
                "description": "Country Code of the caller",
                "configurable": true,
                "valueType": "string",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type": "CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            },
            {
                "name": "Originating Number",
                "description": "Phone number of the caller",
                "configurable": true,
                "valueType": "string",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type": "CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            },
            {
                "name": "Destination Country Code",
                "description": "Country Code of the recipient",
                "configurable": true,
                "valueType": "string",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type": "CharacteristicSpecification",
                "extensible": true,
```


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```
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            },
            {
                "name": "Destination Number",
                "description": "Phone number of the recipient",
                "configurable": true,
                "valueType": "string",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type": "CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            },
            {
                "name": "Duration",
                "description": "Duration of the call in seconds",
                "configurable": true,
                "valueType": "number",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type": "CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            },
            {
                "name": "startDateTime",
                "description": "Start date and start time of the call",
                "configurable": true,
                "valueType": "dateTime",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type": "CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            },
            {
                "name": "endDateTime",
                "description": "End date and end time of the call",
                "configurable": true,
                "valueType": "dateTime",
                "minCardinality": 1,
                "maxCardinality": 1,
                "@type": "CharacteristicSpecification",
                "extensible": true,
                "isUnique": true,
                "validFor": {
                    "startDateTime": "2020-09-23T16:42:23Z"
                }
            }
        ],
        "@type": "ServiceUsageSpecification",
```


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```
        "serviceUsageSpecRelationship": [
            {
                "id": "8b64-0cfd1a91r0xx",
                "href":
"https://api.csp.com/serviceUsageManagement/v5/serviceUsageSpecification/8b64-
0cfd1a91r0xx",
                "name": "voip-Usage-Specification",
                "@type": "ServiceUsageSpecRelationship"
            }
        ]
    }
```

Deletes a ServiceUsageSpecification

`DELETE /serviceUsageSpecification/{id}`

Description

This operation deletes a ServiceUsageSpecification entity.

Usage samples

Here's an example of a request for deleting a specific ServiceUsageSpecification.

Request

```
DELETE /serviceUsageSpecification/8b64-0cfd1a90c9ed
Content-Type: application/json
```

Response

```
204

{
    "id": "8b64-0cfd1a90c9ed",
    "href":
"https://api.csp.com/serviceUsageManagement/v5/serviceUsageSpecification/8b64-
0cfd1a90c9ed",
    "@type": "ServiceUsageSpecification"
}
```

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API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

POST /hub

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request

```
POST /api/hub

Accept: application/json

{
    "callback": "http://in.listener.com"
}
```

Response

```
201

Content-Type: application/json

Location: /api/hub/42

{
    "id":"42",
    "callback": "http://in.listener.com",
    "query":""
}
```

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Unregister listener

DELETE /hub/{id}

Description

Clears the communication endpoint address that was set by creating the Hub..

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request

```
DELETE /api/hub/42

Accept: application/json
```

Response

```
204
```

Publish Event to listener

POST /client/listener

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

Usage Samples

Here's an example of a notification received by the listener. In this example "EVENT TYPE" should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

Request

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```
POST /client/listener

Accept: application/json

{
    "event": {
        EVENT BODY
    },
    "eventType": "EVENT_TYPE"
}
```

Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

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Acknowledgements

Release History

Release Number	Date	Release led by:	Description
Version 5.0.0	October 2023	Abdul Majid Hussain Telstra abdul.majidhussain@team.telstra.com	Upgrade to V5 design patterns and OAS3
Version 5.0.0	March 2023	Varun Nair Telstra varun.nair@team.telstra.com	Upgrade to V5 design patterns and OAS3
Version 5.0.1	Sep 2024	Sergei Lukin Deutsche Telekon sergei.lukin@telekom.de	Align with Resource Usage and Product Usage
Version 5.0.1	14 Oct 2024	Sergei Lukin Deutsche Telekon sergei.lukin@telekom.de	release

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